

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY													
TEACHING & EXAMINATION SCHEME FOR B TECH (CBCS) PROGRAMME IN AIML (2022-23 Admitted Batch)													
Sem	Course Code	Course Title	Teaching Scheme						Examination Scheme				
			Contact Hours				Credit		Theory		Practical		Total
			Theory	Practical	Tutorial	Project	Theory	Practical	Internal	External	Internal	External	
FY Sem-1	MA143	Engineering Mathematics-I	4	0	1	0	4	0	30	70	0	0	100
	CE143	Computer Concepts & Programming	3	4	0	0	3	2	30	70	50	50	200
	EE145	Basics of Electronics & Electrical Engineering	3	2	0	0	3	1	30	70	25	25	150
	IT144	ICT Workshop	0	2	0	0	0	1	0	0	25	25	50
	PY142	Engineering Physics - I	0	2	0	0	0	2	0	0	50	50	100
	FS101A	Foundation Course on Mathematics and Physics	0	2	0	0	0	2	0	0	50	50	100
	HS 101.02 A/ HS 102.02 A	Communicative English / Communicative English - I	2		0	0	2	0	0	0	30	70	100
							12	8					800
FY Sem-2	MA144	Engineering Mathematics-II	4	0	1	0	4	0	30	70	0	0	100
	CE144	Object Oriented Programming with C++	3	4	0	0	3	2	30	70	50	50	200
	ME145	Elements of Engineering	3	2	1	0	3	1	30	70	25	25	150
	FS101A	Foundation Course on Chemistry and Biology	0	2	0	0	0	2	0	0	50	50	100
	PY143	Engineering Physics - II					0	2			50	50	100
	CL144A	Environmental Sciences	0	2	0	0	0	2	0	0	30	70	100
	CSE101	Digital Electronics	2	2	0	0	2	1	30	70	25	25	150
	HS103.02 A / HS201.02 A	Communicative English - II /Painting / Photography / Sculpting / Pottery and Ceramic Arts / Media and Grapic Design / Art and Craft / Fashion Designing /	2		0	0	2	0	0	0	30	70	100
							14	10					1000
Sem	Course Code	Course Title	Teaching Scheme					Examination Scheme					
			Contact Hours				Credit	Theory		Practical		Total	
			Theory	Practical	Tutorial	Total		Internal	External	Internal	External		
SY Sem-3	AIML204	Introduction to Artificial Intelligence	3	2	0	5	4	30	70	25	25	150	
	AIML205	Data Analysis using Python	2	2	0	4	3	30	70	25	25	150	
	AIML201	Java Programming	2	2	0	4	3	30	70	25	25	150	
	AIML202	Data Communication & Networking	3	2	0	5	4	30	70	25	25	150	
	AIML203	Data Structures & Algorithms	3	4	0	7	5	30	70	50	50	200	
	XXXXX	University Elective - I	2		0	2	2	-	-	30	70	100	

	HS121.02 A	Creativity Problem Solving and Innovation	0	2	0	2	2	-	-	30	70	100
			13	16	0	29	23	150	350	210	290	1000
SY Sem-4	AIML209	Probabilistic Modelling and Reasoning with Python	3	2	0	5	4	30	70	25	25	150
	AIML210	R Programming for Data Science And Data Analysis	2	2	0	4	3	30	70	25	25	150
	AIML206	Database Management System	3	2	0	5	4	30	70	25	25	150
	AIML207	Design & Analysis of Algorithms	3	2	0	5	4	30	70	25	25	150
	AIML208	Operating System	3	2	0	5	4	30	70	25	25	150
	XXXXX	Univeristy Elective - II	0	2	0	2	2	-	-	30	70	100
	AIML211	PROJECT - I	0	2	0	2	2	-	-	50	50	100
	HS111.02 A	Human Values and Professional Ethics	0	2	0	2	2	-	-	30	70	100
			14	16	0	30	25	150	350	235	315	1050
Sem	Course Code	Course Title	Teaching Scheme					Examination Scheme				
			Contact Hours				Credit	Theory		Practical		Total
			Theory	Practical	Tutorial	Total		Internal	External	Internal	External	
TY Sem-5	AIML303	Machine Learning and Pattern Recognition	2	2	0	4	3	30	70	25	25	150
	AIML304	DATA ANALYTICS AND VISUALIZATION	2	2	0	4	3	30	70	25	25	150
	AIML301	Cryptography & Network Security	3	2	0	5	4	30	70	25	25	150
	AIML302	Full Stack Web Development	0	4	0	4	2	0	0	50	50	100
		Programme Elective-I	2	2	0	4	3	30	70	50	50	200
	AIML305	PROJECT - II	0	2	0	2	2	-	-	50	50	100
	AIML306	Summer Internship-I			0	0	3	0	0	75	75	150
	HS131.02 A	Communication and Soft Skills	0	2	0	2	2	0	0	30	70	100
		Program Elective - II (RM/CP)	2			2	2					
			9	16	0	27	24	120	280	330	370	1100
TY Sem-6	AIML313	Neural Networks	2	2	0	4	3	30	70	25	25	150
	AIML314	Advance Machine Learning with python	2	2	0	4	3	30	70	25	25	150
	AIML311	Software Engineering	3	2	0	5	4	30	70	25	25	150
	AIML312	Cloud Computing	3	2	0	5	4	30	70	25	25	150
		Programme Elective-III	2	2	0	4	3	30	70	25	25	150

	AIML315	PROJECT - III	0	2	0	2	2	-	-	50	50	100
	HS132.02 A	Contributory Personality Development	0	2	0	2	2	0	0	30	70	100
		Program Elective - IV (RM/CP)	2			2	2					
			12	14	0	28	21	150	350	205	245	950
Final Year Sem-7		Programme Elective-V	4	4	0	8	6	25	75	50	50	200
		Programme Elective-VI	4	4	0	8	6	25	75	50	50	200
		Minor Project	0	20	0	20	10	-	-	125	175	300
		Summer Internship -II	0	0	0	0	3	-	-	75	75	150
			8	28	0	36	25	50	150	300	350	850
Final Year Sem-8		Major Project	0	36	0	36	18	-	-	250	350	600
			0	36	0	36	18	-	-	250	350	600
							180					

LIST OF ELECTIVE SUBJECTS FOR B TECH PROGRAMME IN AIML

ELECTIVES	Code	Programme Elective - I	Code	Programme Elective - III	Code	Programme Elective - V/VI
	AIML307	Advanced Java programming	OCAIML3001	Blockchain Technologies (NPTEL)		Big Data Computing
	AIML308	Mobile Application Development	OCAIML3002	Internet of Things (NPTEL)		Modern AI
	AIML309	Embedded Systems	OCAIML3003	Compiler Construction (NPTEL)		Reinforcement Learning
	AIML310	.NET Core Programming	AIML316	Image Processing		Natural Language Processing
* Programme Elective Subjects will be offered based on availablility of courses on NPTEL/SWAYAM Platform. * Programme Elective Subjects Names may vary based on course offered on NPTEL/ SWAYAM Platform.						Computational Complexity
						Algorithmic Game Theory
						Deep Learning for Computer Vision
						Ethical Hacking
						Social Network Analysis

LIST OF UNIVERSITY ELECTIVE SUBJECTS FOR B TECH PROGRAMME

VES	Code	University Elective - I (UE - I)	Code	University Elective - II (UE - II)
	EC281.01	Introduction to MATLAB Programming	EC282.01	Prototyping Electronics with Arduino
	CE281.01	Art of Programming	CE282.01	Web Designing
	CL281.01	Environmental Sustainability and Climate Change	CL282.01	Basics of Environmental Impact Assessment
	EE284	Python Programming	EE287 EE288	Matlab Programming Maintenance of household apparatus
	IT283	Web Designing and UI/UX	IT284	Data Visualization

ELECTIV

ME281.01	Engineering Drawing	ME282.01	Material Science
PH233	Fundamentals of Packaging	PH238	Cosmetics in daily life
PD260	Basic Laboratory Techniques	NR261	Life Style Diseases & Management
NR251	First Aid & Life Support	PT192	Occupational Health & Ergonomics
PT191	Health Promotion and Fitness	CA225	Programming the Internet
CA224	Introduction to Web Designing	BM241	Health Care Management
BM231	Banking and Insurance		